



APEXPLANET SOFTWARE PVT. LTD.

ApexPlanet Sales & Customer Analytics

From Raw Data to Revenue Strategy:

A 60-day data analytics capstone turning 1,000 transactions into a ₹139.4M growth roadmap.

Data Wrangling → EDA & BI → RFM Segmentation → Statistical Validation → Strategy



DATA



EDA



RFM



VALIDATE

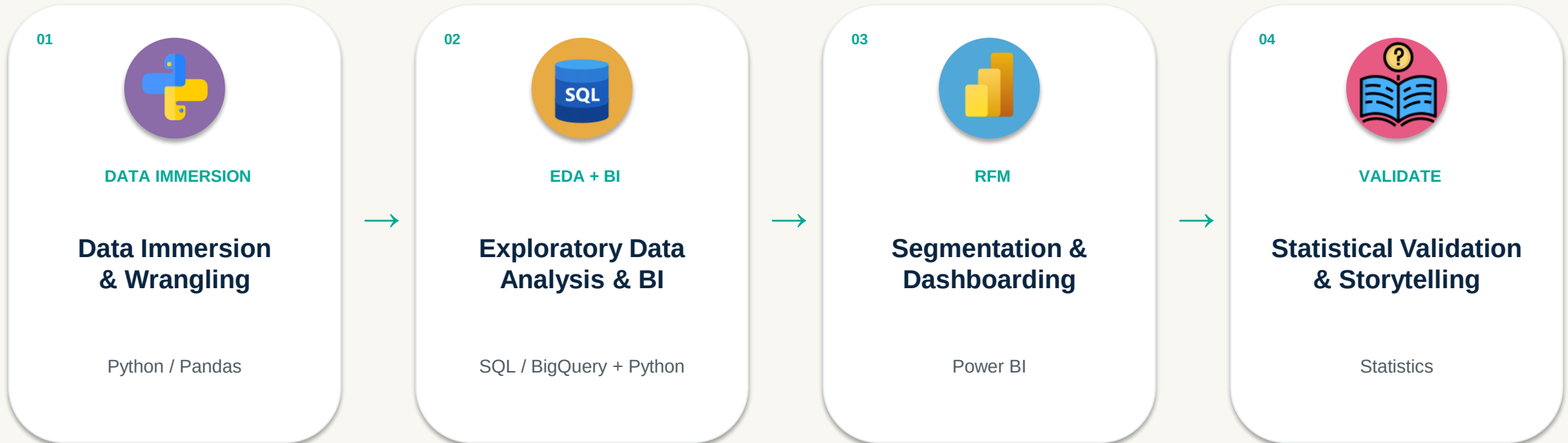


STRATEGY



Project Objective & Roadmap

Four structured phases turned messy transactional data into a validated, actionable growth strategy.



CREDIBILITY BACKBONE

Clean → Explore → Segment → Stress-test → Recommend

Executive Summary — The Business in Numbers

₹139.4M in revenue across 947 customers reveals a healthy but concentration-exposed and re-engagement-ready customer base.

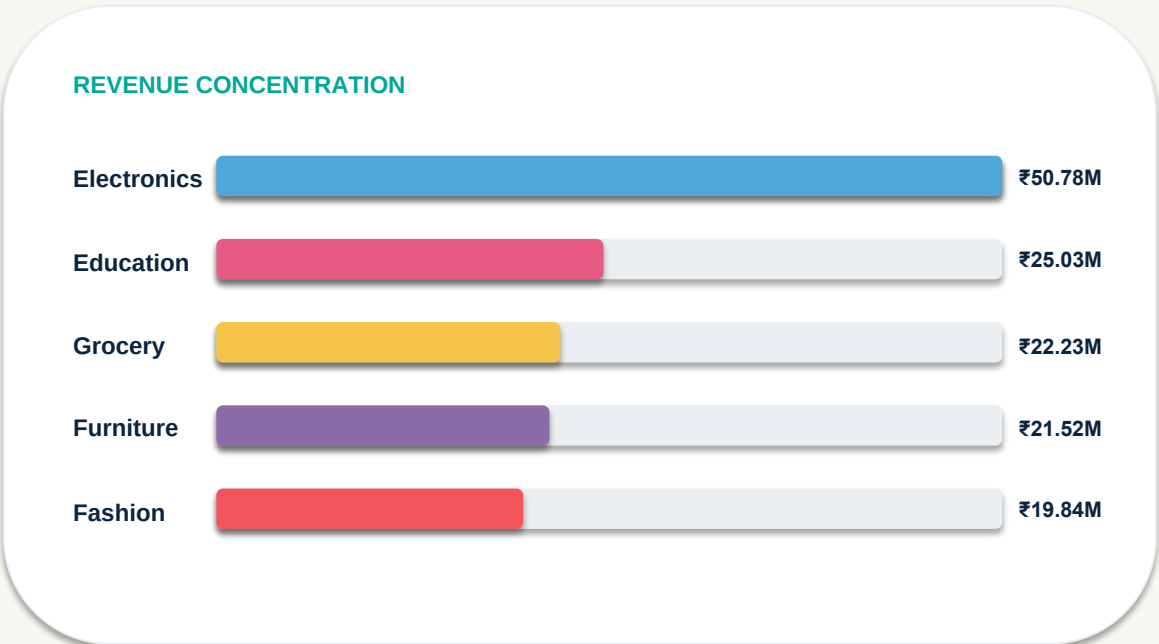


THE BUSINESS SIGNAL

Electronics = 36.4% of total revenue

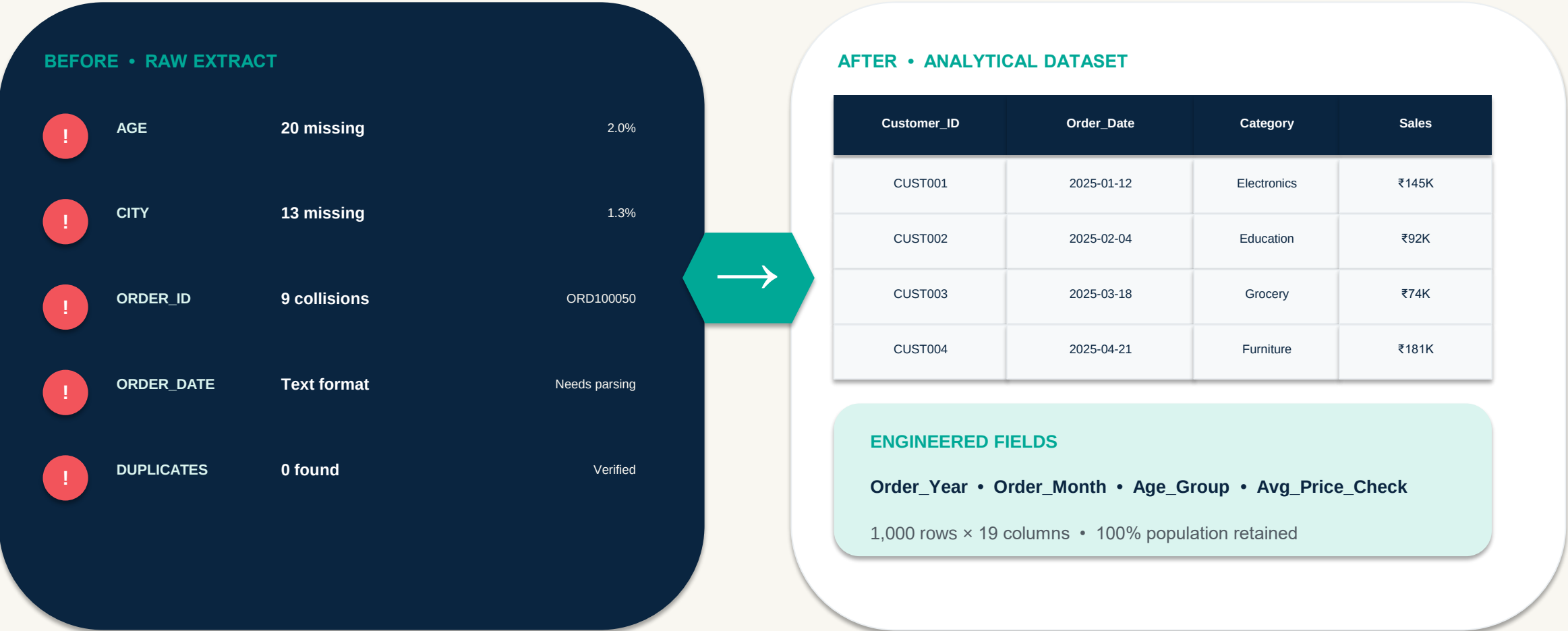
₹50.78M largest single category

476 customers across At Risk + Lost segments represent ₹41.6M in exposed historical revenue.



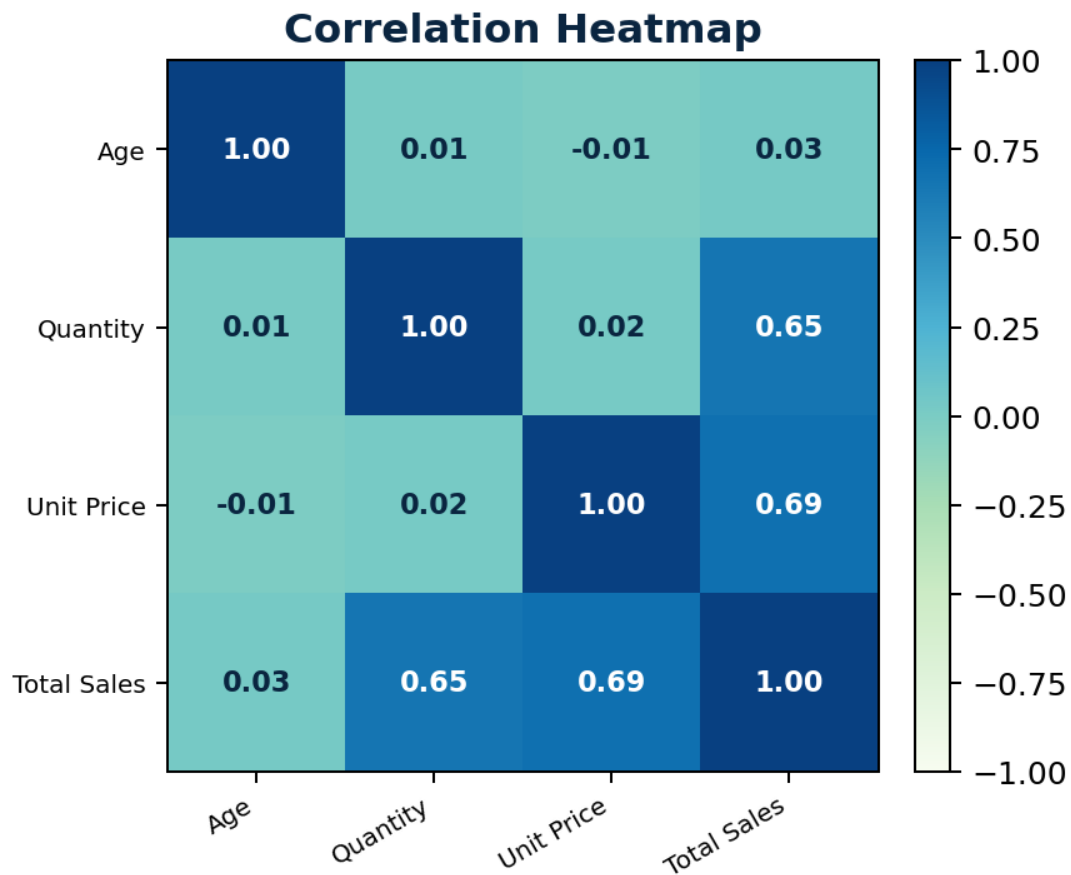
Building a Trustworthy Data Foundation

Systematic cleaning turned a 1,000×12 raw extract into a fully auditable 1,000×19 analytical dataset with zero records lost.



What Actually Drives a Transaction's Value

Transaction value is driven by price and volume, not by customer age or gender.



THE TWO STRONGEST SIGNALS

$$r = 0.69$$

Unit Price ↔ Total Sales

$$r = 0.65$$

Quantity ↔ Total Sales

Age: $|r| \leq 0.03$

Quantity ↔ Unit Price: $r = 0.02$

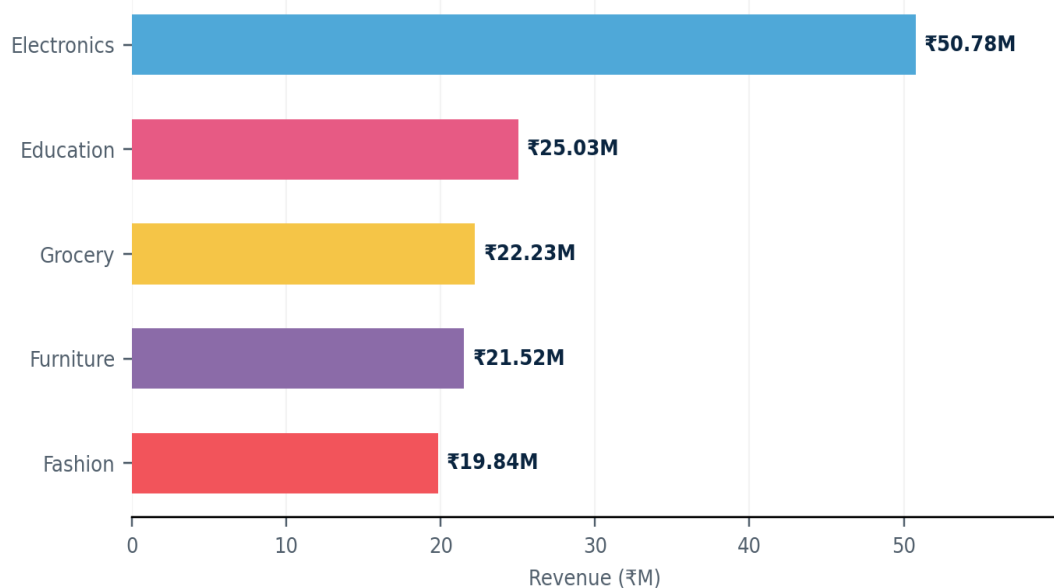
- Pricing and basket-size mechanics matter more than demographic targeting for AOV.
- Bundles, thresholds and tiered offers are therefore more data-supported levers.

Revenue Concentration Across Category and Geography

Revenue is broad-based across 8 cities but concentrated in one category — Electronics.

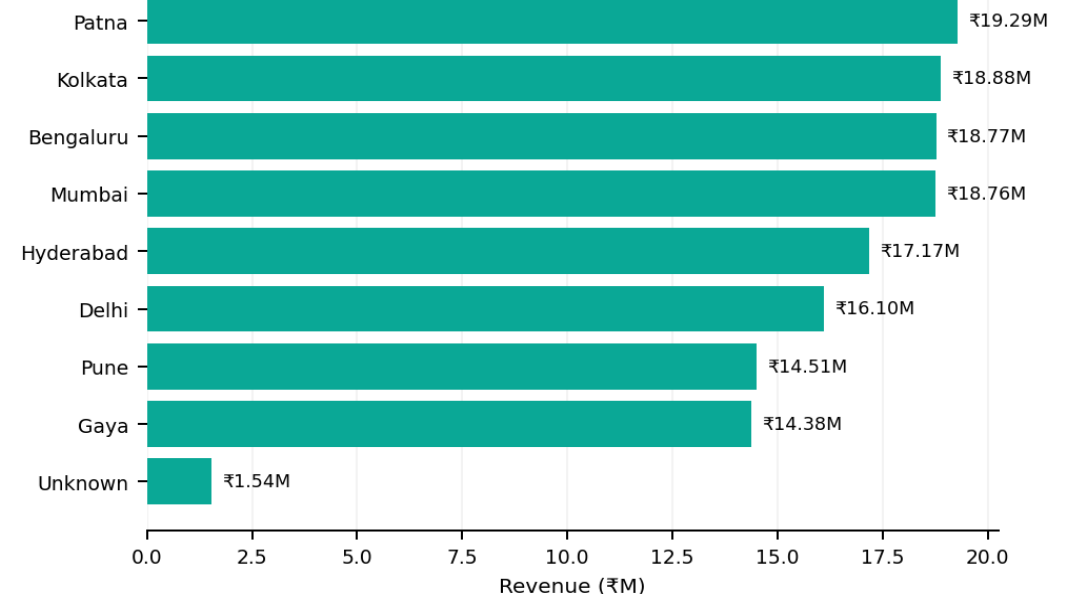
CATEGORY REVENUE

Revenue by Category



GEOGRAPHIC REVENUE

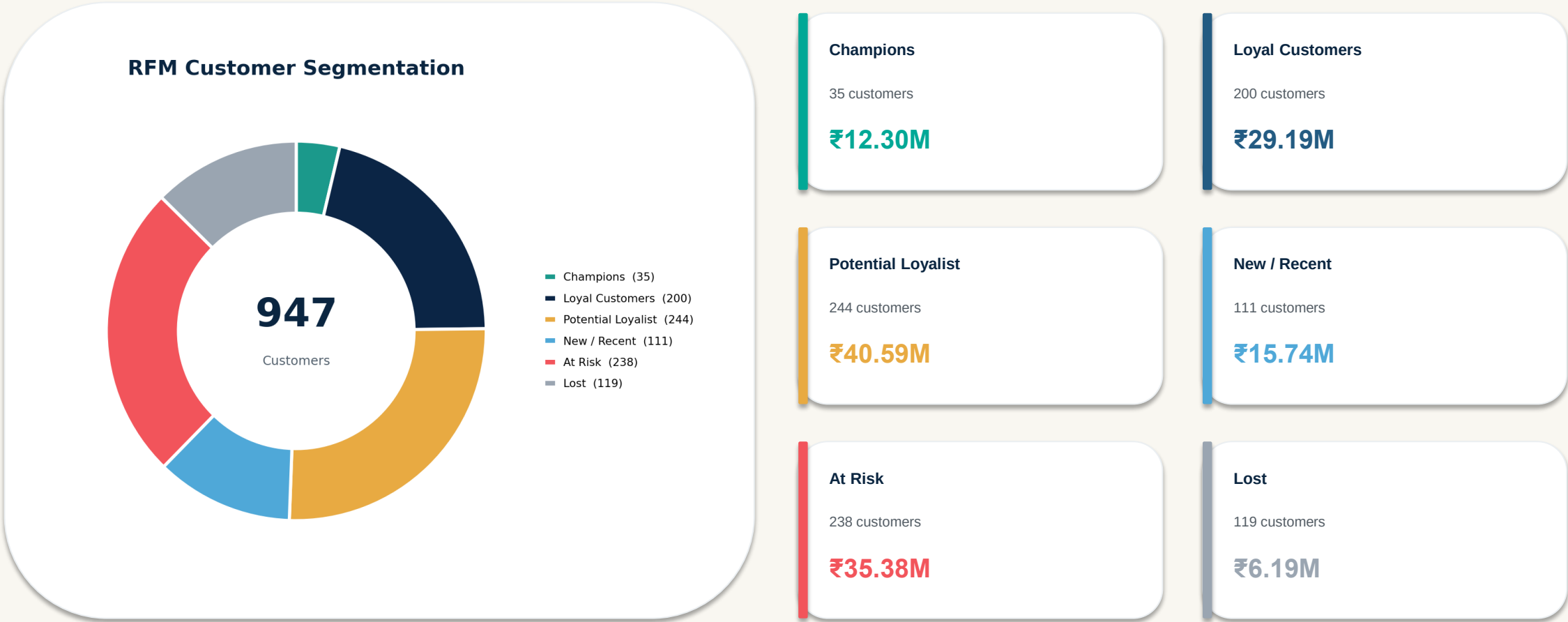
Revenue by City



Electronics ₹50.78M = 36.4% • No single city exceeds 14% of revenue

Segmenting 947 Customers by Recency, Frequency & Monetary Value

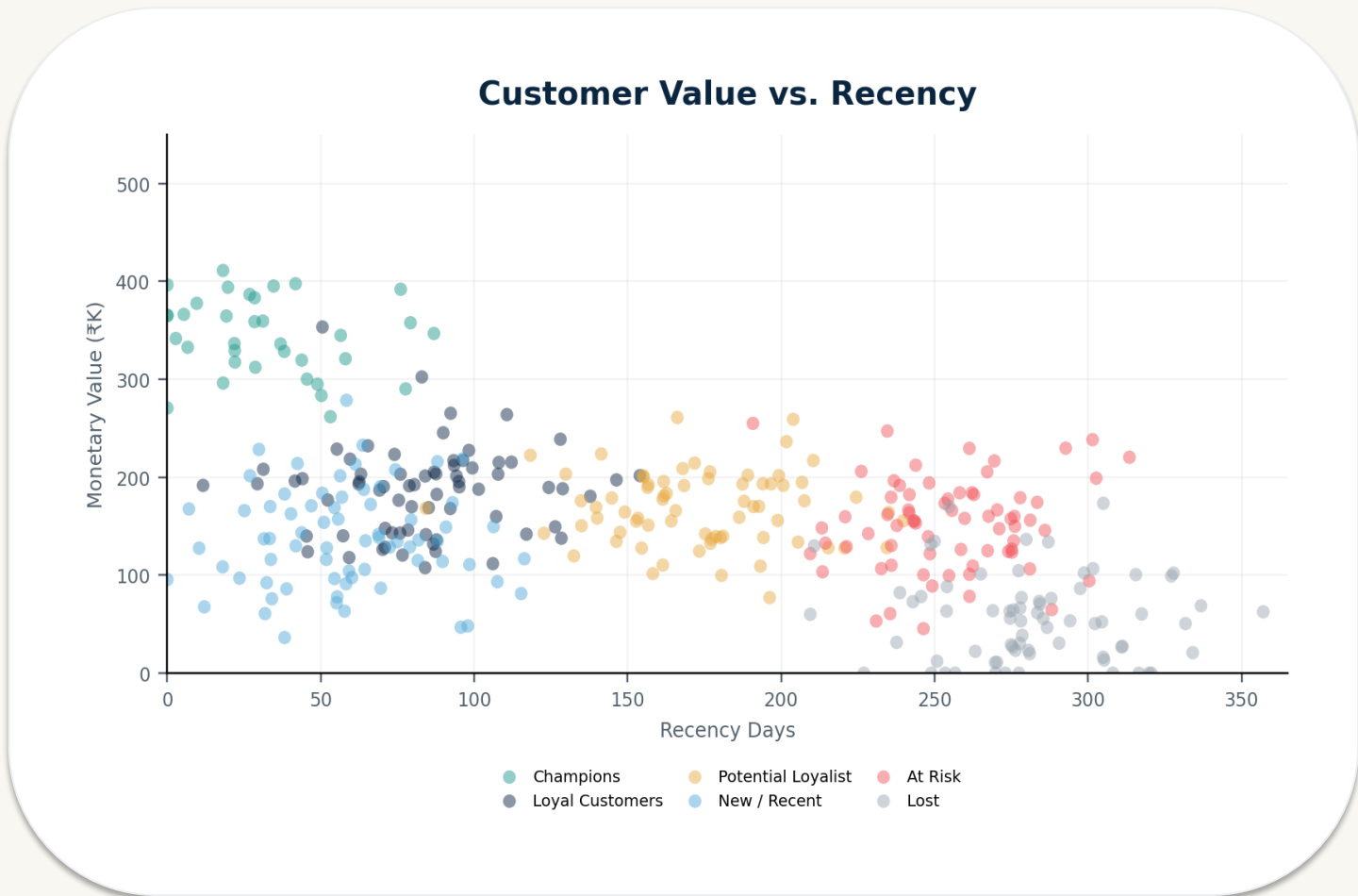
RFM segmentation splits the customer base into six behaviorally distinct groups, each requiring a different strategy.



Average RFM Score: 7.50 • Scale: 3–12

Where the Value Sits — and Where It's at Risk

₹41.6M in customer value sits in the At-Risk and Lost segments — the single largest business exposure in the dataset.



VALUE EXPOSURE

₹41.57M

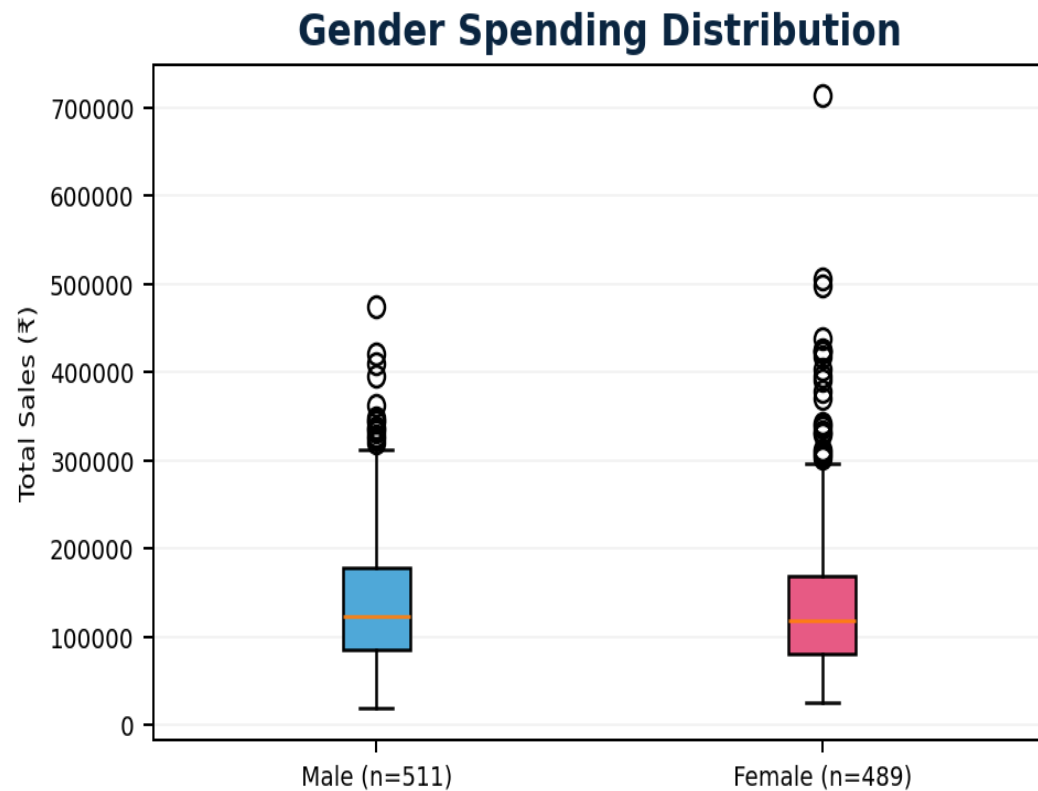
At Risk + Lost

357 customers historical value at risk

AT RISK	238 customers
CHAMPIONS	35 customers
POTENTIAL LOYALIST	₹40.6M pool
FREQUENCY	Mostly one-time

Testing the Assumptions: Gender-Based Spending

No statistically significant difference in spending between male and female customers.



INDEPENDENT TWO-SAMPLE T-TEST

t = 0.6820

p = 0.495389

NOT SIGNIFICANT

Male mean

₹141,807

Female mean

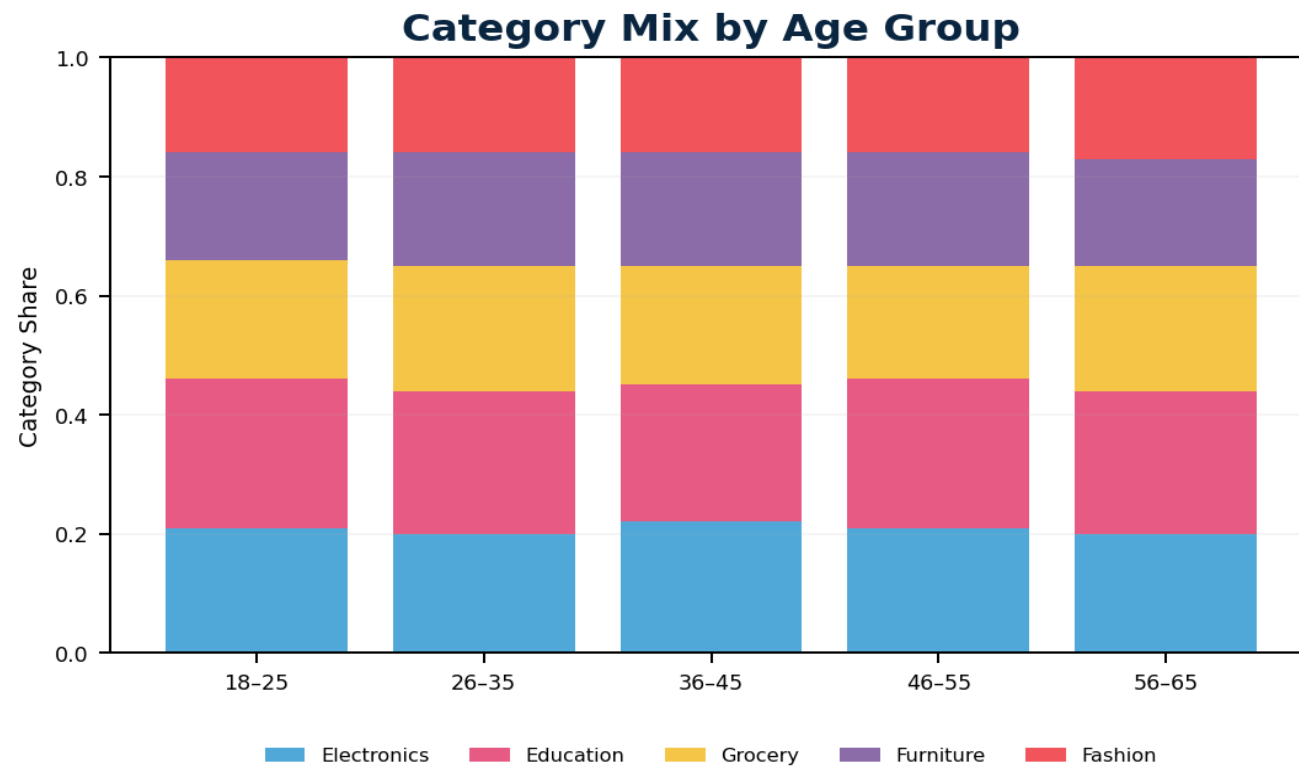
₹136,883

95% CI for difference: -₹9,214.68 to ₹19,062.94

Decision: Fail to reject H_0 at $\alpha = 0.05$

Testing the Assumptions: Do Age Groups Prefer Different Categories?

Product category preference is independent of customer age group.



CHI-SQUARE TEST OF INDEPENDENCE

$$\chi^2 = 10.5361$$

df = 16

$$p = 0.837174$$

NOT SIGNIFICANT

$$\text{Cramér's } V = 0.0513$$

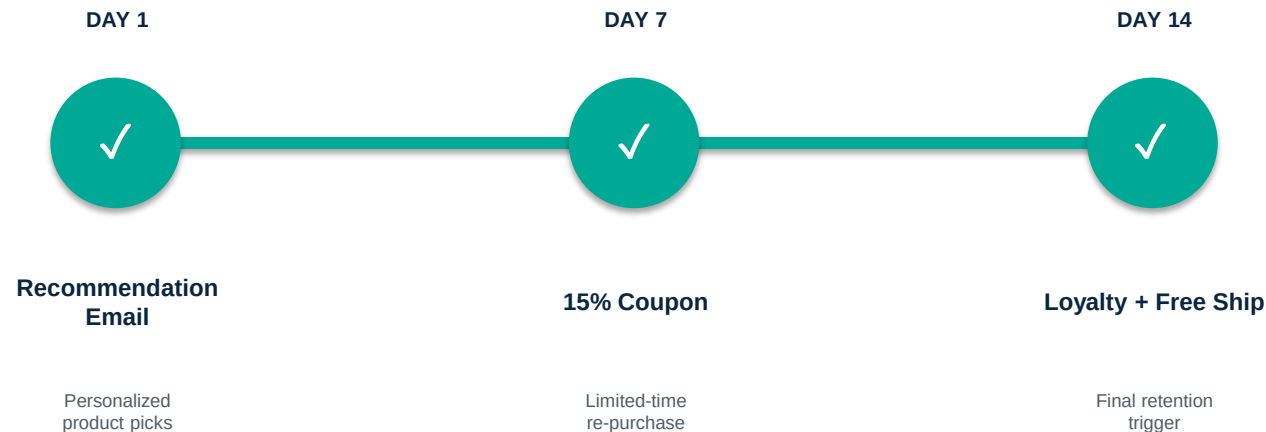
Negligible association

Decision: Fail to reject H_0

Immediate Action: Win-Back the At-Risk Segment

A 14-day, 3-touch win-back sequence can recover ₹3.6M–5.4M from 238 At-Risk customers at a 3:1–5:1 ROI.

14-DAY WIN-BACK SEQUENCE



Target: 238 customers • Exposure: ₹35.4M • Investment: ₹2M–3M

EXPECTED RETURN

₹3.6M–5.4M

incremental revenue

10–15%

expected re-purchase rate

3:1–5:1 ROI

High spend $\geq$ ₹200K: phone + ₹500 off

Low spend $<$ ₹50K: email + ₹100 off

Short-Term: Lock In Champions, Upsell Loyal Customers

Protecting 35 Champions and upgrading 200 Loyal Customers can add ₹2.7M–4.7M in incremental annual spend.

TRACK A

Champions VIP

35 customers • ₹12.3M historical value

- Platinum tier
- 5% cash back
- Dedicated account manager
- Early product access

+₹1.2M–1.8M

ROI 2:1

TRACK B

Loyal Cross-Sell

200 customers • ₹29.2M historical value

- 1% / 3% / 5% cash-back tiers
- Birthday triggers
- SMS engagement
- Cross-category offers

+₹1.5M–2.9M

ROI 1.5:1–2:1

Combined investment: ~₹2.8M–3.6M • Focus: retention + lifetime value

Medium-Term: Convert First-Time Buyers into Repeat Customers

A low-cost, automated second-purchase nudge for New/Recent customers delivers the highest ROI of any initiative — up to 130:1.

SECOND-PURCHASE CONVERSION FUNNEL

111 NEW / RECENT CUSTOMERS

DAY 7 • EMAIL + 20% OFF

DAY 14 / 21 • SMS

22–33 SECOND ORDERS

LOW-COST / HIGH-RETURN

₹3.1M–4.6M

expected revenue

₹35K–45K

estimated campaign cost

ROI 85:1–130:1

Offer: ₹500–₹1K off
Minimum order: ₹5K
Conversion window: 30 days

Strategic: Reactivate Potential Loyalists, De-Risk Electronics Concentration

Reactivating dormant Potential Loyalists and diversifying away from Electronics protects ₹40.6M in value and reduces category concentration risk.

TRACK A • POTENTIAL LOYALIST REACTIVATION



244 customers • ₹40.6M historical value • 6.8+ months dormant

Loyalty relaunch + retroactive points + 48-hour early access

₹8.6M–10.2M

ROI 2.6:1–3.1:1

TRACK B • ELECTRONICS BUNDLING



Electronics + Education bundles • Cross-category recommendations

Target Electronics share: 36.4% → 32%

₹2.5M–5M

incremental revenue

~₹4.8M–5.8M combined investment

Four Initiatives, One Sequenced Strategy

Sequenced correctly, these four initiatives can generate ₹16.8M–25.1M in incremental revenue against ~₹10.6M–13.7M in combined investment.



PARALLELIZE: New/Recent automation + At-Risk win-back • THEN: Loyalty expansion • FINALLY: Dormant reactivation + category diversification

THANK YOU

From data wrangling to a validated,
ROI-ranked growth strategy.

Questions welcome.

CAPSTONE DELIVERABLES

Clean analytical dataset • SQL + Python EDA • Power BI dashboards • RFM model • Statistical validation

Brojo Mohan Dutta

Data Analytics Intern • ApexPlanet Software Pvt. Ltd.

Full technical workflow, SQL, Python notebooks and Power BI dashboard available on request

